FTL Syria – AI for Climate

Applied Machine Learning Team Challenge

1. Team allocation

Challenge	Assigned teams
Problem 1 – Crop Yield Prediction	Groups 1, 2, 3, 4
Problem 2 – Drought Risk Classification	Groups 5, 6, 7,8
Problem 3 – Climate Vulnerability Clustering	Groups 9, 10, 11, 12
Problem 5 – Solar Energy Potential	Groups 13, 14, 15
Problem 4 – Climate Anomaly Detection	Any group

2. The five practical challenges

Problem 1 – Predicting crop yields under climate change

Climate challenge

Climate change affects agricultural productivity through changes in temperature, rainfall, drought frequency and solar radiation. Farmers and policymakers need to understand how these climate variables influence crop production.

Problem statement

Develop a machine learning model that predicts agricultural crop yield based on historical climate and agricultural data.

Research question:

Can historical weather conditions help us predict crop yields, and which machine learning model produces more accurate predictions?

Recommended datasets

1. FAOSTAT – Crop production, harvested area and yield by country and year. Download crop production dataset.
2. NASA POWER – Historical temperature, precipitation and solar radiation. NASA POWER Daily API.

Suggested variables

Variable	Role
Average temperature	Feature
Maximum temperature	Feature
Annual/seasonal rainfall	Feature
Solar radiation	Feature

Country and year	Identification/time features
Crop type	Feature
Crop yield (hg/ha)	Target

Models to implement

- Model A: Decision Tree Regressor
- Model B: XGBoost Regressor

Required evaluation

- MAE – Mean Absolute Error
- RMSE – Root Mean Squared Error
- R^2 – Coefficient of Determination

Expected outcome

Compare the two models and identify which climate variables contribute most to crop yield prediction. Discuss how the findings could support climate-resilient agriculture.

Important: Do not use production volume or harvested area as predictors when predicting yield unless you can justify their availability at prediction time. Avoid mixing future climate information into historical predictions.

Problem 2 – Drought risk prediction and classification

Climate challenge

Drought is one of the most significant climate threats to food security, water availability and agricultural livelihoods.

Problem statement

Develop a machine learning classification system that identifies whether a location is experiencing normal rainfall conditions or drought conditions based on historical meteorological data.

Research question:

Can machine learning identify drought conditions using historical rainfall and temperature patterns?

Recommended datasets

1. NASA POWER – Daily and monthly rainfall and temperature. Access NASA POWER.
2. CHIRPS – Historical rainfall observations and estimates. CHIRPS precipitation data.
3. World Bank Climate Change Knowledge Portal – Historical and projected climate information. Climate data portal.

Suggested variables

Variable	Role
Monthly precipitation	Feature
Average temperature	Feature
Maximum temperature	Feature
Rainfall anomaly	Feature
Previous months' rainfall	Feature
Drought / No drought	Target

Models to implement

- Model A: Decision Tree Classifier
- Model B: XGBoost Classifier

How to create the target variable

If an observed drought classification dataset is unavailable, teams should derive a transparent rainfall-based drought proxy.

For example:

- Normal: seasonal rainfall at or above the 20th percentile of the historical baseline.
- Drought: seasonal rainfall below the 20th percentile.

Required evaluation

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Expected outcome

Identify which model better detects drought conditions, especially drought cases that could otherwise be missed. Explain the consequences of false negatives for early-warning systems.

Problem 3 – Identifying climate-vulnerable regions using clustering

Climate challenge

Different countries and regions experience climate change differently. Some have high exposure to droughts, others face flooding, extreme temperatures or agricultural vulnerability.

Governments and development organisations need to identify regions with similar climate vulnerability profiles to prioritise interventions and resources.

Problem statement

Use unsupervised machine learning to group countries or regions according to their climate exposure, agricultural dependence and socioeconomic vulnerability.

Research question:

Can clustering algorithms identify groups of countries with similar climate vulnerability characteristics?

Recommended datasets

1. World Bank World Development Indicators – Population, agriculture, GDP, rural population and other socioeconomic indicators. World Development Indicators.
2. World Bank Climate Change Knowledge Portal – Historical climate information. Climate data download.
3. FAOSTAT – Agricultural indicators. FAOSTAT database.

Suggested variables

Variable	Description
Average annual temperature	Climate exposure
Annual precipitation	Water availability
Temperature anomaly	Climate change exposure
Agriculture (% of GDP)	Economic dependence
Rural population (%)	Population vulnerability
Agricultural land (%)	Agricultural dependence

Models/techniques to implement

- Technique A: K-Means Clustering
- Technique B: Clustering

Required evaluation

- Silhouette Score
- Davies-Bouldin Index
- Cluster visualisation using PCA
- Interpretation of cluster characteristics

Expected outcome

Identify approximately 3–5 climate vulnerability groups, where justified by the data.

For example:

- Cluster 1: High agricultural dependence and high drought exposure.
- Cluster 2: High temperature exposure and low rainfall.
- Cluster 3: Lower agricultural dependence and greater socioeconomic resilience.

These are illustrative interpretations, not predetermined results. Students must derive their conclusions from their actual clusters.

Problem 4 – Predicting solar energy potential using climate data

Climate challenge

Solar energy is an important renewable energy resource. However, its availability varies according to geographical location, solar radiation, cloud conditions and weather patterns.

Understanding this variability can help governments and energy planners identify suitable locations for solar energy investment.

Problem statement

Develop a machine learning model that estimates daily solar energy potential using historical meteorological variables.

Research question:

Can machine learning predict solar radiation based on meteorological conditions, and which model performs better?

Recommended dataset

NASA POWER – Solar and meteorological data, including daily solar radiation, temperature, wind and humidity.

NASA POWER open dataset.

Suggested variables

Variable	Role
Average temperature	Feature
Maximum temperature	Feature
Relative humidity	Feature
Wind speed	Feature
Cloud amount, where available	Feature
Day of year	Feature
Solar radiation (ALLSKY_SFC_SW_DWN)	Target

Models to implement

- Model A: Decision Tree Regressor
- Model B: XGBoost Regressor

Required evaluation

- MAE
- RMSE
- R^2
- Actual versus predicted solar radiation plot

- Feature importance

Expected outcome

Determine which model provides better solar radiation predictions and discuss the implications for renewable energy planning.

Important: The target is solar irradiation (a measure of solar energy resource), not actual electricity generation from a photovoltaic installation. Students should report the units provided by NASA POWER.

Problem 5 – Detecting climate anomalies and extreme weather events

Climate challenge

Climate change is associated with changes in the frequency and intensity of extreme weather events. Detecting unusual weather patterns can help support climate monitoring and early-warning systems.

Problem statement

Build an anomaly detection system capable of identifying unusual weather patterns in **historical meteorological data.**

Research question:

Can unsupervised machine learning identify unusual temperature and precipitation patterns **without requiring labelled extreme-weather events?**

Recommended datasets

1. NASA POWER – Daily meteorological data. NASA POWER Daily API.
2. NOAA Global Historical Climatology Network Daily – Historical daily weather observations. NOAA GHCN-Daily.

Suggested variables

Variable	Role
Daily maximum temperature	Feature
Daily minimum temperature	Feature
Daily precipitation	Feature
Relative humidity	Feature
Wind speed	Feature
Temperature anomaly	Feature

Models/techniques to implement

- Technique A: Isolation Forest
- Technique B: Local Outlier Factor (LOF)

Required evaluation

Because this is an unsupervised task, accuracy alone is not appropriate.

Students should evaluate:

- Percentage of observations classified as anomalies.
- Agreement between the two techniques.
- Distribution of detected anomalies over time.
- Comparison with historical climate percentiles.
- Manual inspection of selected anomalous events.

Expected outcome

Identify unusual weather events and explain whether they represent potential heatwaves, extreme rainfall, unusually dry periods or other weather anomalies.

Students must distinguish genuine climate extremes from data quality problems and sensor or measurement errors. An anomaly is not automatically evidence of climate change.